Jinglong Xu

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Professional Summary

Radio-Frequency (RF), Analog, and mm-Wave Integrated Circuit (IC) designer and Ph.D. candidate at ETH Zürich with 7+ years of IC design and research experience. Multidisciplinary expertise spans RF/mm-wave IC and system design, advanced packaging, and semiconductor materials and processing. Extensive experience across the complete IC development cycle, from architecture and transistor-level circuit design to EM/layout co-design, PCB/antenna integration, and measurement. Proven silicon validation experience includes 20+ tape-outs with first-pass silicon success across multiple CMOS nodes, SOI, and SiGe technologies. Research and design experience spans frequencies from sub-GHz to D-band, with applications in LPWAN/IoT, 5G NR, 6G, SATCOM, ISAC, and high-speed wireline SoCs for AI data-center interconnects. A strong research track record includes 20+ first-author and co-authored publications in top-tier international journals and conferences, with demonstrated contributions to next-generation wireless and wireline communication systems.

Education

- **Swiss Federal Institute of Technology Zürich (ETH Zürich)** — Switzerland
  Ph.D Candidate — *Oct. 2023 – Present*
  Advisor: *Prof. Hua Wang* ***Research Interests:*** *RF/mm-Wave Integrated Circuits and Systems, Phased Arrays, Power Amplifiers*
  Honors: *Winner of* ***EuMIC 2025*** *Young Engineer Prize*

- **Korea Advanced Institute of Science and Technology (KAIST)** — South Korea
  M.S. – Electrical Engineering; GPA: 91.33/100 — *Sep. 2020 – Aug. 2022*
  Advisor: *Prof. Sang-Gug Lee* ***M.S. Thesis Title:*** *Design of CMOS Receivers for 5G NR and LPWAN Applications*
  Honors: *Recipient of* ***ISSCC 2022*** *Student Travel Grant*

- **Harbin Institute of Technology (HIT)** — China
  B.E. – Material Science and Engineering - Electronic Packaging; Major GPA: 3.76/4.00 *Sep. 2016 – Aug. 2020*
  Honors: *Recipient of the National Scholarship by China Scholarship Council (****CSC****); First Prize of People's Scholarship (2016 & 2017)*

- **University College Dublin (UCD)** — Ireland
  CSC Outstanding Undergraduate Exchange Program; GPA: 3.98/4.20 — *Sep. 2018 – Aug. 2019*
  Major Courses: *Radio-Frequency Electronics* ***(A+)****; Solid-State Electronics* ***(A+)****; Biophysics at the Nanoscale and Nanodevices* ***(A+)****; Nanomaterials* ***(A+)****; Advanced Metals/Materials Processing* ***(A+)****; Physics of Nanomaterials* ***(A+)***

Work Experience

- **RF Technical Consultant – Point2 Technology Inc.** — South Korea
  D-Band (176 GHz) Wireline Transceiver SoC in 90-nm SiGe for an 800 Gb/s Link — *Sep. 2022 – Jun. 2023*
  - **D-Band Receiver**: Designed key receiver sub-blocks, including mixer, clock path, power divider, and output driver.
  - **RX Optimization**: Conducted RX full-chain optimization for PAM-4, including LNA, mixer, LO, and output driver.
  - **Tape-outs**: 90-nm SiGe BiCMOS *Nov. 2022* (High-Band); 90-nm SiGe BiCMOS *Mar. 2023* (Dual-Band)

- **Contract Researcher – KAIST** — South Korea
  Sub-6-GHz CMOS Wireless Receivers for IoT & 5G NR Applications — *Sep. 2022 – Oct. 2023*
  - **Low-Power WuRX**: Led the receiver design for a $0.2 - 1.0$ GHz radio SoC integrating RF, MODEM, and PMU unit.
  - **Duty-Cycling Scheme**: Implemented a novel two-step wake-up scheme for the sub-10-μW duty-cycled WuRX.
  - **Time-Modulated RX**: Designed the DLL and RX for a novel time-divided I/Q switched capacitor RX architecture.
  - **Tape-outs**: 55-nm CMOS *Oct. 2022*; 65-nm CMOS *May. 2023* (Time-Modulated RX); 55-nm CMOS *Sep. 2023*
  - **Related Publications**: [P1], [C7], [J10].

Research Experience

- **RF/cm-Wave Power Amplifiers for Satellite and 6G FR3 Applications** — ETH Zürich
  Ph.D. Candidate — *Oct. 2023 – Present*
  - **Compact Doherty**: Developed Doherty architectures with novel single-footprint transformer combiners and leakage-coupled passive-network modeling at FR3; achieved 25.3 dBm P_{sat} and 29.7% PAE at 6-dB power back-off.
  - **VSWR-Resilience**: Proposed and developed a negative-g_m-boosted source-follower PA for FR3 with inherent VSWR resilience and reconfigurability; implemented a complex LC-neutralized PA at Ka band with 4:1 VSWR-resilience.
  - **Tape-outs**: 22-nm SOI; Led 5^+ PA tape-outs in GF 22FDX+ with first-pass silicon success for array integration.
  - **Related Publications**: [J2], [C2], [C3], [J3], [J4], [C5], [C8], [C9].

- **Phased-Array ICs and Systems for SATCOM and 6G ISAC Applications** — ETH Zürich
  Ph.D. Candidate — *Oct. 2023 – Present*
  - **Phased Arrays**: Developed phased-array transmitters at Ku- and Ka-band integrating VSWR-resilient PA solutions, phase and gain control, digital control, and power management IPs for scalable satellite communication systems.

- **Time-/Frequency-Modulated Arrays**: Developed novel frequency-modulated and time-modulated array architectures for integrated sensing and communication applications and co-apertured full-duplex operation at FR3.
- **Tape-outs**: 22-nm SOI; Completed 5^+ tape-outs in GF 22FDX+, with array-level IC, PCB and antenna integration.
- **Related Publications**: [J11]. *Further measurements and manuscripts in preparation and planned.*

- **Multi-Band Radio SoC with a Sub-10-μW Wake-Up RX for IoT Applications** KAIST
  M.S. Candidate *Aug. 2021 – Oct. 2022*
  - **RX Circuit**: Designed LNA, mixer, baseband PGA, complex IRNF, and BFSK demodulator for the RX signal path.
  - **Duty-Cycling Scheme**: Proposed a novel two-step wake-up duty-cycling scheme for the sub-10-μW wake-up RX.
  - **Taped-out**: 55-nm CMOS; RX architecture verified, achieved -96 dBm sensitivity for 250 kb/s R_{chip} under 300 μW.
  - **Related Publications**: [C6], [J9], [J10].

- **Sub-6-GHz Low Noise Front-End Module for 5G NR Applications** KAIST
  M.S. Candidate *Sep. 2020 – Aug. 2021*
  - **Ultra-Low-Noise LNA**: Designed an $n79$ $_{(4.7GHz)}$ gain-switching LNA with proposed linearity improvements.
  - **Testbench Modeling**: Developed a strong foundation in parasitics-aware RF design, with model-validated testbench.
  - **Taped-out**: 65-nm CMOS; Achieved the best FoM among LNAs at $4-6$ GHz, outperforming commercial benchmarks.
  - **Related Publications**: [C1].

- **Graphite Nanoplatelet-Based Flexible Electronics and Thermal Management** HIT
  Undergraduate Research Assistant *Sep 2017 – Sep. 2020*
  - **Flexible Electronics**: Designed GNP-based flexible strain/pressure sensors with application on soft robotic control.
  - **Thermal Management**: Designed thermal interface structure for soft and high thermal conductivity PCB substrates.
  - **Related Publications**: [J1], [J5], [J6], [J7], [J8].

Skills

- **Languages**: Mandarin (Native), English (Fluent, IELTS 7.5), Korean (Basic), German (Basic, Targeting B1).
- **IC Simulation**: Cadence Virtuoso, EMX, Keysight ADS, Matlab, Simulink, Ansys HFSS, and related EDA tools.
- **IC Evaluation**: Test Automation, NF and Sensitivity, Linearity, PA Power & Modulation, and related RF testing.
- **Soft Skills**: Strong Work Ethic, Communication, Teamwork, Leadership, Time Management, Public Speaking.
- **Hobbies**: Long-Distance Cycling, Bikepacking, Marathon Running, Skiing (Cross-Country & Alpine), Cooking.

Publications

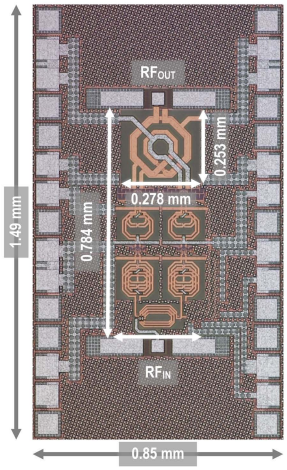
Google Scholar ORCID LinkedIn **J** = Journal **C** = Conference **P** = Patent

- [C5]: **J. Xu**, A. Habib, T.-Y. Huang, N. Villaggi, H. Wang, "A Negative-gm-Boosted Source-Follower Power Amplifier with Inherent VSWR-Resilience and Core-Side Impedance Reconfiguration," ***ISSCC 2027***. ***Under Review***
- [J4]: **J. Xu**, E. Liu, M. Eleraky, T.-Y. Huang, C. Chu, H. Wang, "A Compact Ku-Band FR3 Doherty Power Amplifier with a Stacked Three-Coil Transformer Combiner: Leakage-Coupled Modeling and Schematic-to-EM Synthesis Methodology," ***TMTT 2026***. ***In Preparation***
- [J3]: **J. Xu**, T.-Y. Huang, M. Eleraky, I. Tolaib, B. Abdelmagid, H. Wang, "An Ultra-Compact Series Doherty Power Amplifier with a Single-Footprint Output Combiner for Ku-Band FR3," ***JSSC 2026 Special Issue***. ***Invited Paper***
- [C3]: **J. Xu**, T.-Y. Huang, M. Eleraky, I. Tolaib, H. Wang, "20.1 An Ultra-Compact Asymmetrically Load-Pulled Series Doherty PA in CMOS-SOI with 25.3dBm P_{sat} and 29.7% PAE_{6dB} for Ku-Band 6G FR3," ***ISSCC 2026***. ***Top Ranked, Invited to JSSC 2026 Special Issue***
- [C2]: **J. Xu**, E. Liu, M. Eleraky, T.-Y. Huang, C. Chu, H. Wang, "An Ultra-Compact Ku-Band Doherty PA with a Single-Footprint Triple Two-Turn Asymmetric Combiner for 6G FR3," ***EuMIC 2025***. ***Young Engineer Prize Winner***
- [J2]: **J. Xu**, M. Eleraky, T.-Y. Huang, C. Chu, H. Wang, "A Compact Doubly Neutralized Ku Band Power Amplifier with 39% Peak PAE and 23 dBm Output Power in 22FDX+ EDMOS for 6G FR3," ***MWTL 2025***. ***Invited, IMS 2025 Top-50 Papers***
- [C1]: **J. Xu**, K.-M. Kim, H.-U Mahmood, J. Kim and S.-G Lee, "An n79 Sub-1-dB Noise Figure Highly Linear Variable-Gain LNA Employing Adaptive Imbalanced Bleeding for 5G NR," ***A-SSCC 2022.***
- [J1]: **J. Xu**, H. Wang, T. Ma, Y. Wu, R. Xue, H. Cui, X. Wu, Y. Wang, X. Huang, W. Yao, "A graphite nanoplatelet- based highly sensitive flexible strain sensor," ***Carbon 2020. IF: 11.6***

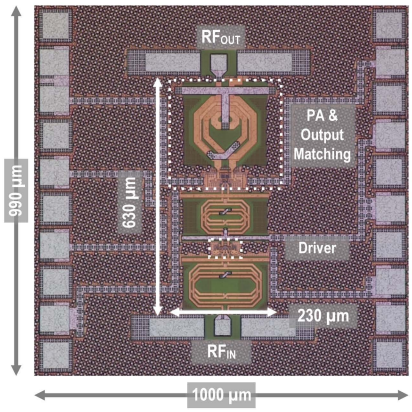
- [J11]: M. Eleraky, A. Novello, G. Cristiano, A. Beam, Y. Liu, J. Zeng, **J. Xu**, B. Lin, J. Nann, H. Liu, M. Saif, N. Villaggi, L. Recchioni, T.-Y. Huang, T. Burger, T. Kleier, H. Wang, "A Thermally Adaptive Ku-Band Array-Scalable Transmitter Incorporating VSWR-Resilient PA for Ground Terminal Phased-Array Applications," ***JSSC 2026.***
- [C9]: F. Svelto, B. Abdelmagid, M. Ghorbanpoor, **J. Xu**, A. Wang, H. Wang, "A 25-GHz Ultra-Compact, VSWR-Resilient Asymmetrical Doherty Power Amplifier with Reconfigurable Inductorless Adaptive Biasing Supporting 12 Gbps 64-QAM," ***ESSERC 2026.***

- **[J10]**: D. Tran, H.-U Mahmood, **J. Xu**, K.-M. Kim, K.-S. Choi, J. Lee, J. Ko, S.-G. Lee, J. Kim, "A Multi-Band Ultra-Low-Power Transceiver with Two-Step Wake-Up Reception," ***SSCL 2026.***
- **[C8]**: C. Chu, **J. Xu**, Y. Liu, J. Zeng, A. Wang, T. Torii, S. Shinjo, K. Yamanaka, H. Wang, "AI-Assisted Template-Seeded Pixelated Design for Multi-Metal-Layer High-Coupling EM Structures: A Ku-Band 6G FR3 PA in 22nm FDX+," ***IMS 2025.***
- **[C7]**: D. Tran, H.-U Mahmood, **J. Xu**, K.-M. Kim, K.-S. Choi, J. Lee, J. Ko, S.-G. Lee, J. Kim. "Multi-band LPWAN Transceiver with Two-step Wake-up Receiver and Switched Capacitor PA Based Transmitter Achieving -125 dBm Sensitivity, 6.7 uW at 1 s Wake-up Latency, and 1Mbps BFSK Modulation," ***A-SSCC 2025.***
- **[C6]**: H.-U Mahmood, K.-M. Kim, D. Tran, **J. Xu**, A. Qahir, J. Ko, J. Kim, S.-G. Lee, K.-S. Choi, "A 24% Efficient, 15.36 dBm Output Power, Multi-Standard Digital Polar Transmitter with 7-bit Phase Interpolator-based BFSK Modulator and 23 dB Sidelobe Suppressed PA for Low-Power Wide Area Networks," ***A-SSCC 2024.***
- **[J9]**: K.-M. Kim, K.-S. Choi, H. Jung, B. Yun, **J. Xu**, J. Ko, S.-G. Lee, "A −124-dBm Sensitivity Interference-Resilient Direct-Conversion Duty-Cycled Wake-Up Receiver," ***JSSC 2023.***
- **[P1]**: S.-G. Lee, K.-M. Kim, J. Ko, K.-S. Choi, **J. Xu**, E.-R. Jeong, "A Duty-Cycling Method for Wake-Up Radios," Korean Domestic Patent Number: 10-2022-0054080, 02 May, 2022.
- **[J8]**: X. Wu, H. Wang, Z. Wang, **J. Xu**, Y. Wu, R. Xue, H. Cui, C. Tian, X. Huang, B. Zhong, "Highly Conductive Thermal Interface Materials with Vertically Aligned Graphite-Nanoplatelet Filler towards: High Power Density Electronic Device Cooling," ***Carbon 2021. IF: 11.6***
- **[J7]**: Y. Du, Y. Xu, W. Zhou, Y. Yu, X. Ma, F. Liu, **J. Xu**, Y. Zhu, "MOF-derived zinc manganese oxide nanosheets with valence-controllable composition for high-performance Li storage," ***Green Energy & Environment 2021. IF: 10.7***
- **[J6]**: X. Wu, Y. Wang, H. Wang, B. Zheng, Y. Wu, R. Xue, **J. Xu**, H. Cui, W. Yao, B. Zhong, X. Huang, "Semi-quantitative orientation control of graphite nanoplatelets in GNP/PU nanocomposite via balancing the effects of gravity and micro-flow field and application in manufacturing heat spreader substrate with excellent thermal conductivity," ***Composites Part A 2021. IF: 8.1***
- **[J5]**: Y. Wang, H. Wang, F. Liu, X. Wu, **J. Xu**, H. Cui, Y. Wu, R. Xue, C. Tian, B. Zheng, W. Yao, "Flexible printed circuit board based on graphene/polyimide composites with excellent thermal conductivity and sandwich structure," ***Composites Part A 2021. IF: 8.1***

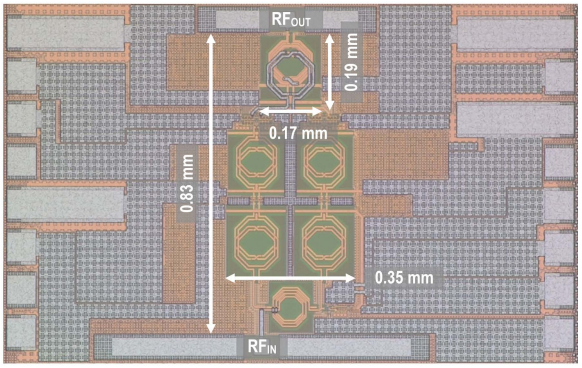
Selected Chip Gallery



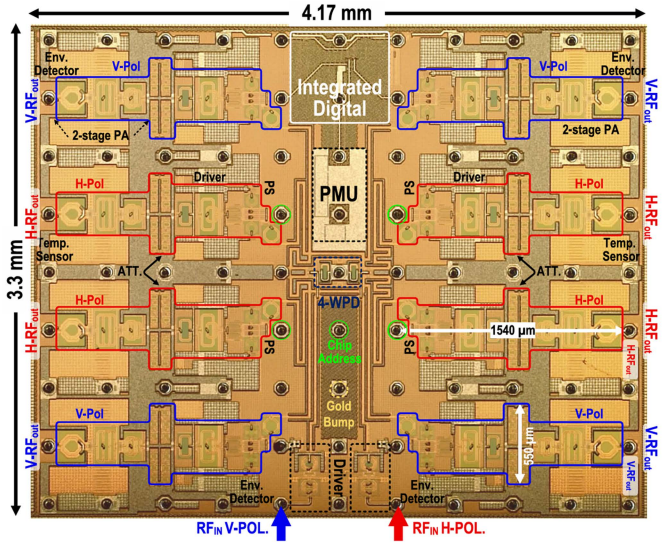
Ku-Band FR3 Compact Series Doherty PA
[C3, ISSCC 2026] [J3, JSSC 2026]



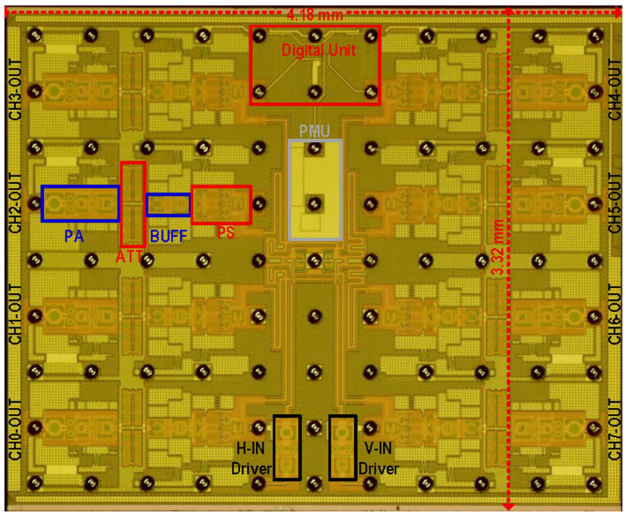
Ku-Band FR3 VSWR-Resilient and Reconfigurable PA
[C5, ISSCC 2027, Under Review]



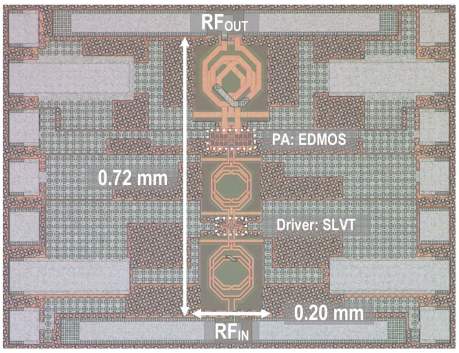
Ku-Band FR3 Compact Parallel Doherty PA
[C2, EuMIC 2025] [J4, TMTT 2026]



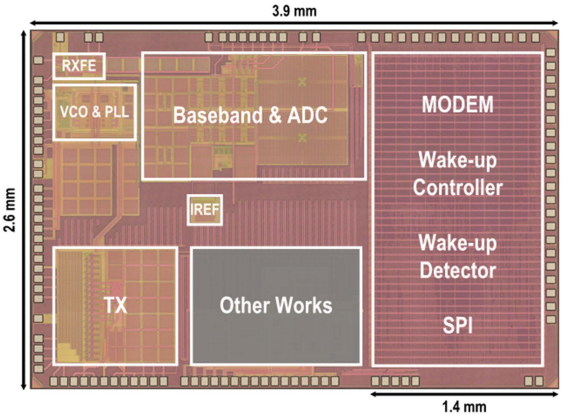
Ku-Band SATCOM 2x4 Transmitter Array
[J11, JSSC 2026]



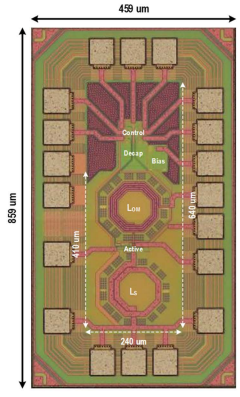
Ka-Band SATCOM 2x4 Transmitter Array
[TMTT 2026, In Prepration]



Ku-Band FR3 High-Power CMOS PA
[J2, IMS 2025 & MWTL 2025]



Multi-Band Sub-GHz LPWAN Transceiver with Ultra-Low-Power WuRX
[C7, A-SSCC 2025] [J10, SSCL 2026]



n79-Band Ultra-Low Noise LNA
[C1, A-SSCC 2023]